

EE3621: Digital Signal Processing

Lecture 26: DSP for Power Systems — Harmonic Analysis & Phasor Estimation (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan (40 Minutes Breakdown)

- 00:00 – 05:00 (5 mins): Power Quality Issues, Harmonics, THD.
 - 05:00 – 12:00 (7 mins): Fourier Analysis of Power Signals, DFT.
 - 12:00 – 17:00 (5 mins): DFT Spectral Leakage and mitigation.
 - 17:00 – 22:00 (5 mins): Phasor Measurement Units (PMUs), TVE.
 - 22:00 – 27:00 (5 mins): Recursive (Sliding) DFT.
 - 27:00 – 32:00 (5 mins): Kalman Filter for Phasor Tracking.
 - 32:00 – 35:00 (3 mins): Protection Relay DSP & Anti-Aliasing.
 - 35:00 – 40:00 (5 mins): Active/Reactive Power from DFT & Checkpoints.
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2 Power Quality Issues

Nonlinear loads (VFDs, UPS, rectifiers) draw non-sinusoidal currents. The Total Harmonic Distortion (THD) quantifies this:

$$THD_V = \frac{\sqrt{\sum_{k=2}^N V_k^2}}{V_1} \quad (1)$$

High THD causes motor overheating, transformer derating, and capacitor bank failures.

3 Fourier Analysis of Power Signals

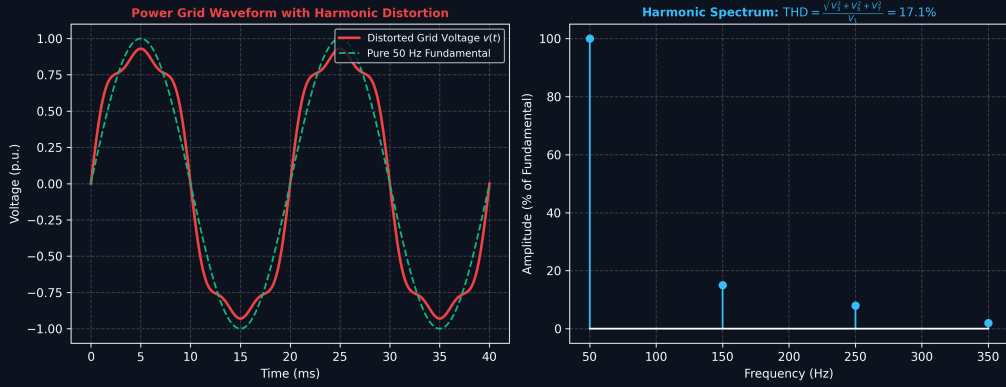


Figure 1: Power Grid Voltage Distortion and Harmonic Spectrum with Total Harmonic Distortion (THD) Calculation.

Sliding Recursive DFT Architecture for Real-Time Grid Phasor Estimation

Recursive Sliding DFT Update Formula:
 $X_n[k] = (X_{n-1}[k] + x[n] - x[n-N]) * W_N^{-k}$
 Computes full new spectrum in only $O(1)$ complex operations per sample!

Phasor Measurement Unit (PMU) Operation in Smart Grids:
 Provides sub-cycle voltage magnitude $|V|$ and phase angle ϕ updates synchronized to GPS clock
 Critical for instantaneous fault detection and wide-area power system stability monitoring!

Figure 2: Recursive Sliding-DFT Architecture for Real-Time GPS-Synchronized Phasor Measurement Units (PMUs).

The fundamental frequency $f_0 = 50/60$ Hz is sampled at f_s , requiring an integer number of samples per cycle $N = f_s/f_0$. The DFT yields complex harmonic phasors:

$$V_k = \frac{2}{N} \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} kn} \quad (2)$$

4 DFT Leakage in Power Systems

When f_0 drifts (e.g., to 49.8 Hz), the N -sample window no longer perfectly spans one cycle. This non-integer cycle lengths cause spectral leakage. Mitigation involves windowing and interpolation-based frequency estimation.

5 Phasor Measurement Units (PMUs)

Synchrophasors are time-synchronized by GPS (1-PPS). The IEEE C37.118 standard requires the Total Vector Error (TVE) to be strictly less than 1%:

$$\text{Phasor} = \frac{V_{peak}}{\sqrt{2}} e^{j\phi} \quad (3)$$

$$\text{TVE} = \sqrt{\frac{(\hat{X}_r - X_r)^2 + (\hat{X}_i - X_i)^2}{X_r^2 + X_i^2}} \times 100\% \quad (4)$$

6 Recursive DFT (Sliding DFT)

To track real-time phasors efficiently, we update the DFT sample-by-sample. Derivation:

$$X_k[n] = \sum_{m=0}^{N-1} x[n-m] W_N^{k(N-1-m)} \quad (5)$$

$$X_k[n] = \left(X_k[n-1] - x[n-N] W_N^0 + x[n] W_N^{kN} \right) W_N^{-k} \quad (6)$$

$$X_k[n] = (X_k[n-1] + x[n] - x[n-N]) e^{j\frac{2\pi}{N}k} \quad (7)$$

This reduces computation from $O(N \log N)$ for a full FFT to $O(1)$ per harmonic per sample.

7 Kalman Filter for Phasor Tracking

The state model uses phasor amplitude and phase as states, and noisy voltage samples as measurements.

$$\mathbf{x}_k = \mathbf{x}_{k-1} + \mathbf{w}_k \quad (8)$$

$$z_k = \mathbf{H}_k \mathbf{x}_k + v_k \quad (9)$$

where $\mathbf{H}_k = [\cos(k\theta_0) \quad -\sin(k\theta_0)]$.

8 Protection Relay DSP

Overcurrent, distance, and differential relays use DFT-based phasor estimation. Anti-aliasing filters are designed to cutoff below the Nyquist frequency. Analog lowpass prototypes for these anti-aliasing filters are shown in the figures below.

9 Active and Reactive Power from DFT

$$P = \frac{1}{2} \sum_k |V_k| |I_k| \cos(\phi_{V_k} - \phi_{I_k}) \quad (10)$$

$$Q = \frac{1}{2} \sum_k |V_k| |I_k| \sin(\phi_{V_k} - \phi_{I_k}) \quad (11)$$

10 Checkpoint Questions

1. **THD Calculation:** Calculate THD given $V_1 = 230$, $V_3 = 40$, $V_5 = 30$.
Answer: $THD_V = \frac{\sqrt{40^2 + 30^2}}{230} = 21.74\%$.
2. **Samples per Cycle:** For $f_s = 2880$ Hz and $f_0 = 60$ Hz, $N = 48$. The 5th harmonic (300 Hz) corresponds to $k = 5$.
3. **Recursive DFT Efficiency:** Recursive updates require just 1 complex multiplication per step, compared to $O(N \log N)$ for a full FFT.